Capstone Lightning Talk #2

Bank Customer Churn Prediction

Predicting which savings customers are about to leave 'before they do'

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Problem Statement & Success Metrics

THE PROBLEM

A retail bank earns from savings deposits. **18.53%** drain their balance and go dormant; the bank finds out after the money is gone.

THE OBJECTIVE

Classify customers as **HIGH RISK** or **LOW RISK** using demographics, branch data, and recent transactions.

SUCCESS METRICS

Metric	Target
Recall (churn)	**≥ 0.70** — catch 7 of every 10 churners
ROC-AUC	**≥ 0.80** — proves the model ranks risk
Precision / F1	Monitored — keeps the call list realistic
~~**Accuracy**~~	NOT USED — a do-nothing model already scores **81%**

Who This Is For

Audience	What they get
Retention team	A ranked list of at-risk customers to call this week
Branch managers	Which branches and customer types are losing deposits
Risk & Finance	How much money is at risk of walking out
Data science team	Maintains and retrains the model

Goals & Objectives

Problem type: CLASSIFICATION

the target **'churn'** is 1 or 0

6 MODELS = 3 MODEL TYPES × 2 VERSIONS

	Model	Version 1	Version 2
A	Logistic Regression (baseline model)	Base	Tuned
B	Random Forest	Base	Tuned
C	XGBoost	Base	Tuned

! A model that predicts "nobody churns" scores 81.47% accuracy — and finds ZERO churners. That is why we measure RECALL, not accuracy.

Evaluation: Recall · ROC-AUC · Precision · F1 · Confusion matrix

About the Data

28,382
CUSTOMERS

21
COLUMNS

18.53%
CHURN RATE

DATA DICTIONARY (GROUPED)

Group	Columns	Examples
Customer	6	age, gender, occupation, dependents, vintage
Location	2	city, branch_code
Balances	6	current, previous month, previous 2 quarters
Transactions	4	credits and debits, this month and last
Engineered (new)	2	no_last_txn, days_since_txn

0 duplicates · **0 rows deleted** **during cleaning**

Key Finding: The Two-Month Drain

Churn becomes visible in the debit pattern before the balance disappears.

When	Stayed	Churned	Signal
Balance, 2 quarters ago	**3,383**	**3,206**	almost identical
Balance, last quarter	**3,508**	**3,721**	churners held MORE
Money out, last month	**43**	**838**	**19x higher**
Money out, this month	**22**	**1,120**	**51x higher**
Balance now	**3,643**	**1,541**	collapsed to under half

> "Churn is not a sudden event. It is a visible drain over about two months."

So the bank should act when debits spike — not after the balance is gone.

EDA for Modelling

CLEANING DONE

- ✓ **3,223** hidden nulls — stored as the text **NaT**
- ✓ **806** customers with an impossible age under **18**
- ✓ **dependents** values up to **52** — capped at **6**

FINDINGS THAT SHAPE THE MODEL

Finding	Decision
Money columns skewed	Log transform — correlation rises **0.048 → 0.236**
6 balance columns correlate up to **0.994**	Keep 1–2; replace the rest with change features
Chi-Square: all **3** categorical features significant	Keep all three — occupation strongest

Outliers were kept, not deleted. In banking the outlier IS the high-value customer — the exact person the model exists to protect.

Progress Report

✓ **DONE: data cleaned · EDA · churn pattern found · modelling plan ready**

CHALLENGES

Challenge	How we handle it
Raw correlations very weak (best 0.073)	Engineered feature stronger
Imbalanced target 18.53%	**class_weight** , SMOTE
branch_code has 3,185 codes	Frequency encoding or group rare codes

NEXT — LIGHTNING TALK #3

1. Build the engineered features
2. Run and compare all 6 models
3. Tune the threshold to reach recall 0.70
4. Deliver the best model